

Mohsin Ali Mohammed

San Diego, CA | +1 (619) 273-7627 | mohammedmohsin5431@gmail.com

Portfolio | LinkedIn | GitHub | Numin: numin.trade | Quant Decision Labs

Applied AI engineer building production LLM evaluation infrastructure, agent systems, retrieval pipelines and fine-tuning workflows. Empirical ML researcher with work accepted at EMNLP 2026 (Main Conference) and prior work in CALCS, HICSS and the AAI Student Abstract and Poster Program.

EXPERIENCE

Telaeris Inc

Applied AI / Software Engineer (Full-time)

May 2023 - Present

San Diego, CA

- Improved task accuracy by 20% relative and reduced hallucination rate by 30% relative through structured prompting, RAG grounding and supervised fine-tuning, with stage-by-stage ablations to attribute gains.
- Built and deployed LangGraph and LangChain agent workflows across OpenAI, Claude and Gemini, integrating 10+ tools and APIs with structured outputs, guardrails, routing, memory, human review, automated recovery and LangSmith tracing.
- Deployed 5 model-serving microservices on AWS/EKS using FastAPI, Docker, Kubernetes and CI/CD, with vLLM/SGLang inference at 1,000+ requests/day; cut median latency 25% and cost per request 20%.
- Designed a 1,500+ example LLM evaluation dataset for credential intake, exception handling and policy Q&A, defining label schemas, annotation guidelines and gold sets; froze baselines before model development.
- Built an evaluation harness scoring 2,000+ responses on accuracy, groundedness, hallucination rate and tool-call correctness; validated LLM judges against human spot-audits and tested 200+ adversarial probes.
- Raised hybrid-search Hit@10 from 0.74 to 0.83 through embedding retrieval, query understanding, metadata enrichment and Pinecone-based ranking, validated by A/B test.

Rupeek Fintech Pvt Ltd

Machine Learning Engineer

Aug 2021 - Aug 2022

Bangalore, India

- Built forecasting and risk pipelines for 700+ loan portfolios on Databricks/PySpark; deployed XGBoost with drift-based retraining, SHAP explainability and ongoing performance monitoring under model governance.
- Developed anomaly detection with Isolation Forest and autoencoders for fraud and risk monitoring; optimized U-Net, YOLOv3 and ONNX computer-vision models, cutting inference latency from 13s to 4s.

PUBLICATIONS

- When Structure Is Hidden: Measuring and Diagnosing the Compositional Gap in LLMs. EMNLP 2026 (Main Conference), accepted. [Project/code](#)
- On the Feasibility of Vision-Language Models for Time-Series Classification. HICSS 2026. [Paper](#)
- CONFLATOR: Incorporating Switching Point based Rotatory Positional Encodings for Code-Mixed Language Modeling. CALCS 2023 workshop proceedings. [Paper](#)
- PESTO: Switching Point Based Dynamic and Relative Positional Encoding for Code-Mixed Languages. AAI 2022, Student Abstract. [Paper](#)

ENTREPRENEURSHIP & RESEARCH

Quant Decision Labs / Numin

Founder, Numin / Lead Researcher

Jan 2025 - Present

San Diego, CA

- Shipped a multi-agent debate platform producing versioned, evidence-grounded theses from real-time market data, using FastAPI, Postgres + pgvector, Redis, Docker and CI/CD.
- Built live tick and L2 depth capture for NQ/ES futures and BTC/ETH, preserving exchange and receipt timestamps with gap/reset checks, bounded recording and backup jobs.
- Acquired and audited 2,880 hourly Parquet files (34 GB) of BTC/ETH depth and trades; built snapshot-anchored replay across hour boundaries and flagged sequence gaps, resets and invalid executions for exclusion.
- Built reproducible point-in-time datasets with availability timestamps and SHA-256 manifests; froze trial grids, block bootstrap, hierarchical FDR, power floors and purge-and-embargo validation before scoring.
- Reported negative NQ results: 4 direction tests failed to beat the reference forecast (BSS -0.115 to -0.094), with 0 of 34 order-flow cells showing lift; limited positive skill appeared in volatility expansion (+0.048) and level contact (+0.057).
- Built a unit-tested deterministic Python control layer the model cannot override, enforcing risk floors, confidence and coherence checks, and a latching daily-loss halt, with rule-level decision audit logs.

SUBMITTED MANUSCRIPTS

Identity Drift Without Locality Drift: Structured Routing Reorganization Under Continual Adaptation.

Submitted to ICLR 2027. In OLMoE-1B-7B, full-router continual adaptation increased expert-identity drift 1.7x while structured expert substitutions stayed stable to +/-0.002 across 3 seeds, under marginal-, size- and position-controlled nulls.

From Discovery to Replication: Stress-Testing Empirical Laws in Continual Learning.

Submitted to ICLR 2027. Pre-registered a cross-scale replication benchmark: across 600 discovery episodes from 3 independent datasets, naive held-out selection produced 98% non-transferable empirical laws. Built ScaleTrust, a frozen trust/abstain protocol calibrated on planted ground truth.

EDUCATION

San Diego State University

M.S. in Computational Data Science

Aug 2022 - May 2024

San Diego, CA

Indian Institute of Information Technology Sri City

B.Tech in Computer Science and Engineering

Aug 2018 - Jun 2022

Sri City, India

SKILLS

AI / ML: LLMs, RAG, agentic AI, multi-agent systems, tool/function calling, LangGraph, LangChain, supervised fine-tuning (LoRA/SFT), sparse Mixture-of-Experts, transformers, computer vision

Evaluation: LLM-as-judge, eval harness design, dataset curation and annotation design, adversarial testing, ablations, calibration, selective prediction, hierarchical FDR, block bootstrap, walk-forward validation, point-in-time data, A/B testing

ML / Serving: PyTorch, scikit-learn, XGBoost, ONNX, vLLM, SGLang, FastAPI, Docker, Kubernetes, AWS/EKS, CI/CD

Languages / Data: Python, SQL, TypeScript, PostgreSQL, pgvector, Pinecone, Redis, Databricks, Spark/PySpark, Databento, Polygon, FRED/ALFRED